An Algorithmic and Complexity Analysis of the Wide-Scan Geometric Resonance Factorization Method

1. Executive Assessment: Viability of the "Wide-Scan Geometric Resonance" Method

This report provides a technical analysis of the "Wide-Scan Geometric Resonance" (WSGR) method, a novel, classical (non-quantum) algorithm for integer factorization developed by independent researcher Dionisio Lopez (zfifteen). The method reframes the $N=pq$ factorization problem as an exercise in Digital Signal Processing (DSP), where the goal is to detect a "resonance peak" within a 2D parameter space.

The core methodology postulates that the factor p can be estimated by a bespoke equation, $\hat p(k,m)=\exp!((\ln N-2\pi m/k)/2)$. This equation defines a signal landscape over the two parameters k (a continuous "frequency" variable) and m (a discrete "mode" integer). The algorithm's "Wide-Scan" strategy involves a deterministic sweep of this 2D space, notably using a Golden-Ratio Quasi-Monte Carlo (QMC) sampler for the k-dimension to ensure even coverage. A Dirichlet kernel, a standard tool in Fourier analysis, is then employed as a "spectral gate" or filter to identify the precise coordinates of a resonance peak, which in turn reveals the factors.

The novelty of the WSGR method lies not in its individual components—QMC samplers and DSP kernels are standard tools—but in its *conceptual framework*. It uniquely combines an unproven *ansatz* from analytic number theory with a practical, brute-force DSP search paradigm. The very term "Geometric Resonance" is a neologism, a metaphorical application of a term borrowed from the disparate fields of rigorous dynamical systems (concerning Cantor sets) [1] and speculative physics.[3]

Despite a reported success in factoring a 127-bit semiprime in 128 seconds, a rigorous analysis of the method's stated scaling hurdles indicates that the WSGR algorithm is almost certainly in the **Exponential (EXP) complexity class**. Its cryptographic viability is fatally undermined by two fundamental scaling issues:

1. **Exponential Sampling Density:** The "peak tightening" hurdle is a direct manifestation of the Nyquist-Shannon sampling theorem. As the input N (and thus its bit-length $L = \log N$) grows, the "frequency" of the target signal in the parameter space increases. This causes the target peak to narrow *exponentially* (e.g., peak width $\Delta k \propto

N^{-c}). To guarantee detection of this peak, the sampling density in the k-dimension must therefore increase exponentially, requiring $O(e^{cL})$ samples.

2. **Polynomial Search Space:** The algorithm's other parameters, including the search range for the m-dimension and the computational cost of each check (the "precision budget"), appear to scale at least polynomially with L. The researcher's use of a *constant* m-range (e.g., [-180, +180]) appears to be a special-case artifact from factoring a balanced (RSA-like) number where $p \approx q$. This range is mathematically indefensible for a *general* factorization algorithm.

In conclusion, the WSGR method is a fascinating and highly original *numerical curiosity*. It serves as a proof-of-concept that a DSP-based analogy can successfully factor specific, small semiprimes. However, its fundamental scaling properties are exponential, rendering it cryptographically irrelevant as a threat to modern (e.g., 2048-bit) semiprimes. It is *not* in the same class as sub-exponential classical sieves (like the General Number Field Sieve (GNFS)) [4] and is infinitely less efficient than polynomial-time quantum methods (like Shor's Algorithm).[5]

2. Deconstruction of the Core "Geometric Resonance" Analogy

This section deconstructs the theoretical and terminological foundations of the WSGR method, focusing on the origin of its name and its central, unproven *ansatz*.

2.1 Tracing the Terminus: "Geometric Resonance"

The provided research confirms that "Geometric Resonance" is *not* a standard term in computational number theory. Its use by the researcher is a neologism, appearing to draw metaphorical support from two distinct and unrelated fields.

1. **Source 1 (Rigorous): Dynamical Systems.** A body of rigorous mathematical work by Peres, Shmerkin, and others uses the term "geometric resonance" in the context of geometric measure theory.[1] Here, it describes a precise technical condition for the arithmetic sum of two self-similar Cantor sets (K, K'). "Geometric resonance" occurs if the Hausdorff dimension of their sum is strictly less than the sum of their individual dimensions ($\dim(K+sK') < \dim(K)+\dim(K')$). This, in turn, implies an "algebraic resonance," where the contraction ratios of the sets are all powers of a common base $r < 1$.[1] This is a deep, formal, but highly specialized concept with no apparent link to integer factorization.
2. **Source 2 (Speculative): Physics.** A separate, non-rigorous usage of "Geometric Resonance" appears in a collection of preprints on speculative physics.[8] This framework attempts to unify field theories, cosmology (e.g., "CMB Quadrupole-Octupole Alignment" [12]), and nuclear physics, positing that mass, charge, and other phenomena

are "coherent expressions of confined geometric resonance".[10]

A key clue connects the WSGR method to the *speculative physics* branch. The researcher's algorithm makes a specific, non-obvious engineering choice: a **Golden-Ratio QMC** sampler. One of the speculative physics papers [3] explicitly proposes that the proton is stabilized by "internal geometric resonance, likely governed by the **Golden Ratio (ϕ)**."

The simultaneous use of the term "Geometric Resonance" and a "Golden-Ratio" sampler creates a strong link. This suggests the researcher is *not* building upon the rigorous Cantor set theorems of Peres/Shmerkin.[1] Instead, the WSGR algorithm appears to be constructed from the *speculative physics analogy* [3], treating the semiprime N as a "resonant structure" that can be "probed" using tools (like the Golden Ratio) considered fundamental within that speculative framework. This confirms the method is based on a *physical metaphor*, not a derived mathematical theorem.

2.2 The Central Ansatz: Deconstructing $\hat p(k,m)=\exp!((\ln N-2\pi m/k)/2)$

This equation is the entire theoretical core of the method. It is the *ansatz* (a postulated, unproven equation) that defines the "signal" the algorithm searches for. As its derivation is not provided, its mathematical basis must be reverse-engineered.

1. **Rewrite the Equation:** The equation is $\hat p(k,m) = \exp((\ln N - 2\pi m/k)/2)$. Taking the natural logarithm gives: $\ln \hat p = (\ln N - 2\pi m/k) / 2$.
2. **Simplify:** This can be rewritten as $\ln \hat p = (\ln N)/2 - \pi m/k$.
3. **Introduce Factor Properties:** We know $N=pq$, so $\ln N = \ln p + \ln q$. The geometric mean of the factors is $\sqrt{N}$, and $\ln \sqrt{N} = (\ln N)/2$.
4. Isolate the Postulate: Let us assume the estimate $\hat p$ is one of the factors, p. Substituting the known properties into the equation from Step 2 yields:
   $\ln p = (\ln p + \ln q)/2 - \pi m/k$
   $2 \ln p = \ln p + \ln q - 2\pi m/k$
   $\ln p - \ln q = -2\pi m/k$
   $\ln(p/q) = -2\pi m/k$

This derivation reveals the researcher's core *postulate*: **The logarithmic ratio of the factors ($\ln(p/q)$) is assumed to be a "resonant" or "quantized" value, directly proportional to an integer m (the "mode") and inversely proportional to a continuous parameter k (the "frequency").**

The presence of 2π strongly implies this is an analogy to phase in signal processing, as seen in Euler's formula ($e^{i\theta}$).[14] The researcher has *equated* a static number-theoretic property (the ratio of factors) with a dynamic DSP property (a phase/frequency value m/k).

This *ansatz* is what transforms factorization into a 2D search problem: if one can find the correct pair (k,m) that "resonates," one can solve for p using the original equation. The 127-bit success is the *empirical evidence* that for at least *some* N, this *ansatz* holds and a

corresponding signal peak can be found. While the link between logarithms of primes and "frequency" or "phase" is conceptually related to ideas in analytic number theory [15], this explicit formulation appears to be the researcher's unique and unproven contribution.

2.3 From "Precision Snipe" to "Wide-Scan"

The researcher's notes indicate an initial "precision snipe" approach failed, forcing a pivot to the "Wide-Scan" (coverage over guesswork). This pivot was a logical and necessary consequence of the chosen *ansatz*.

1. **Failure 1: Trivial "Central Mode".** The researcher's formula for the central mode, $m_0=\mathrm{nint}!(k(\ln N-2\ln\sqrt N)/(2\pi))$, is a trivial identity. Since $2\ln\sqrt N = \ln N$, the expression collapses: $m_0 = \mathrm{nint}!(k(\ln N - \ln N)/(2\pi)) = 0$. This indicates the formula was ill-posed for its intended use as a starting point for a numerical optimization.
2. Failure 2: Extreme Sensitivity. The researcher correctly diagnosed the second failure: extreme sensitivity of the ansatz itself, given as $|\frac{d\hat p}{dm}| \propto \hat p$. A simple check of the derivative confirms this. Let $\hat p(m) = \exp(C - \pi m/k)$, where $C = (\ln N)/2$. The derivative with respect to m is:
   $\frac{d\hat p}{dm} = (-\pi/k) \cdot \exp(C - \pi m/k) = (-\pi/k) \cdot \hat p$

This trivial derivative has a profound implication: the search landscape is not a smooth "bowl" suitable for gradient-descent optimization. It is an *exponentially steep* "canyon" or "needle." The problem is *ill-conditioned*. If the "correct" m is an integer (e.g., 107), being off by even a small *fractional* amount (e.g., 107.1) results in an *enormous* error in the factor estimate $\hat p$.

The researcher thus correctly concluded that an optimization-based "snipe" was impossible. The pivot to a "Wide-Scan" was not a preference but a *necessity* forced by the mathematical properties of the $\hat p(k,m)$ *ansatz*. The problem had to be reformulated from one of *optimization* ("find the root") to one of *search* ("scan all discrete possibilities").

3. Comparative Analysis: A Novel Niche in Factorization

The WSGR method is classical, deterministic (if the QMC is used as a deterministic sequence), and numerical. It bears no mechanistic resemblance to the dominant families of factorization algorithms.

3.1 Mechanistic Distinctions

- **vs. Shor's Algorithm:** Both methods use Fourier analysis, but to fundamentally different ends. Shor's Algorithm uses the **Quantum Fourier Transform (QFT)** to solve a *different problem*: finding the *period* r of the modular exponentiation function $f(x) = a^x \pmod N$.[5] Finding this period r allows factorization with high probability via $\gcd(a^{r/2} \pm 1, N)$. This relies on quantum superposition and phase estimation.[5] WSGR, by contrast, uses a *classical* DSP filter (a **Dirichlet kernel**) to find a *peak* in a *postulated* 2D signal space. Shor's is a "period-finding" algorithm; WSGR is a "peak-finding" algorithm.
- **vs. Schnorr's Method (Lattice-Based):** These methods are from entirely different mathematical domains. Schnorr's method is *algebraic* and *geometric*, using lattice reduction (like LLL) to find short vectors in a lattice that reveal factoring relations.[18] WSGR is *analytic* and *numerical*, involving no lattices or algebraic geometry.
- **vs. Sieves (GNFS/QS):** Sieves like the General Number Field Sieve (GNFS) and Quadratic Sieve (QS) are *combinatorial* algorithms. They work by finding "smooth" numbers (numbers that factor into small primes) that satisfy the congruence $x^2 \equiv y^2 \pmod N$.[4] WSGR is a numerical search and does not sieve for smooth numbers. Sieves are the *de facto* standard for classical factorization because their complexity is **Sub-Exponential (SUBEXP)**.[4] As will be shown, WSGR's complexity appears to be **Exponential (EXP)**, making it asymptotically less efficient.
- **vs. Other Analytical Methods:** Fourier analysis is a standard tool in *analytic number theory* [20], but it is primarily used in a *non-constructive* manner (e.g., to prove theorems about the distribution of primes). The WSGR method is a rare attempt at a *constructive* analytical algorithm: using DSP techniques to *find* the specific factors of a *specific* N.

3.2 Mechanistic Comparison of Factorization Algorithms

The following table contextualizes the WSGR method by comparing its core mechanism and properties against established factorization algorithms.

Algorithm	Core Methodology	Mathematical Domain	Complexity	Quantum/Classical
Wide-Scan Geometric Resonance	**DSP Peak-Finding:** Searches a 2D parameter space (k,m) for a "resonance" peak defined by a postulated *ansatz* $\hat p(k,m)$.	Analytic Number Theory / DSP	**Exponential (EXP)** (Inferred)	Classical
Shor's Algorithm	**Quantum**	Quantum	**Polynomial**	Quantum

	Period-Finding: Uses the QFT to find the period r of $a^x \pmod N$. [5]	Computation / Number Theory	**(BQP)** ($O(L^3)$)	
Schnorr's Method	**Lattice Reduction:** Finds short vectors in a lattice that correspond to factoring relations. [18]	Algebraic Geometry / Lattices	**(Heuristic)** Classical	Classical
General Number Field Sieve (GNFS)	**Combinatorial Sieving:** Finds "smooth" numbers x, y satisfying $x^2 \equiv y^2 \pmod N$. [4]	Combinatorics / Algebraic Number Theory	**Sub-Exponential (SUBEXP)**	Classical
Elliptic Curve Method (ECM)	**Group Order Finding:** Exploits random elliptic curves, hoping to find one with a "smooth" group order $\pmod p$.	Algebraic Geometry / Elliptic Curves	**Sub-Exponential (SUBEXP)** (Heuristic)	Classical

4. An Expert Review of the DSP and QMC Toolkit

The WSGR method is an engineering pipeline: a Sampler (QMC) feeds candidates to a Filter (Dirichlet). The researcher's specific choices for these components are technically sound, yet reveal the method's trade-offs and future scaling plans.

4.1 The "Station-Surfer" (Golden-Ratio QMC Sampler)

The algorithm uses a "Golden-Ratio QMC (low-discrepancy) sampler" to sweep the k parameter range (e.g., [0.25, 0.45]). The stated rationale is to provide *even, deterministic coverage* and avoid two common pitfalls:

1. **"Clumps/Gaps"**: A known problem with standard pseudorandom (Monte Carlo) samplers.
2. **"Aliasing"**: A known problem with simple grid-based searches.

This rationale is perfectly sound. Low-discrepancy sequences (also called quasirandom) are *designed* to fill a space with maximum uniformity, avoiding the "clusters of points" and "gaps" that plague pseudorandom sequences.[22] In numerical integration, this superior uniformity allows QMC methods to achieve an error rate that scales as $O(N^{-1} (\log N)^d)$, asymptotically much better than the $O(N^{-1/2})$ scaling of Monte Carlo methods.[23] The Golden-Section (or Golden-Ratio) search is a well-known and robust technique for 1D optimization and search.[25]

However, a subtle analysis reveals this choice may be sub-optimal. The WSGR problem is a 2D search (m, k). The researcher's implementation uses a 1D QMC (Golden Ratio) for the k-dimension while stepping through the m-dimension with simple integers. This is a "swept 1D QMC," not a true 2D QMC sequence. Research on low-discrepancy sequences indicates that while the Golden Ratio (ϕ_1) is optimal for 1D, the optimal constant for 2D sequences is the "plastic constant" (ϕ_2).[27] A true 2D QMC sequence (such as a Sobol or Halton sequence [22]) would provide more efficient and uniform coverage of the 2D (m, k) space. This suggests the current implementation, while effective, is not fully optimized.

4.2 The "Spectral Gate" (Dirichlet Kernel Filter)

The researcher uses a Dirichlet kernel ($D_J(\theta)$) with a small $J=6$ and a high threshold (0.92) as a "spectral gate" to find peaks. The rationale is that it is the "natural lens" for a signal assumed to be a finite Fourier sum.

This is technically correct. The Dirichlet kernel is the mathematical kernel that perfectly reconstructs a finite Fourier series from its samples; its convolution with a sampled signal *is* the truncated Fourier approximation.[28]

However, in mainstream DSP, the Dirichlet kernel is considered a *poor* filter for spectral analysis. Its main lobe is very narrow, but it suffers from:

1. **Very High Sidelobes:** The first sidelobe is only ~13dB down from the main peak.
2. **Slow Decay:** Sidelobes decay slowly ($1/f$), leading to "spectral leakage," which can create false peaks.[30]

This is why practical DSP applications use *windowing functions* (e.g., Hann, Hamming) to trade a slightly wider main lobe for *dramatically* suppressed sidelobes. The researcher's success with this "bad" filter is explained by two factors: 1) The *ansatz* signal is a clean, mathematical construct, not a noisy real-world signal. 2) The high threshold (0.92) is a brute-force method to *reject* all but the main lobe, effectively ignoring the sidelobe problem.

The researcher's "scaling lab" plan confirms this advanced understanding. The plan proposes a **Two-stage (Fejér + Dirichlet) kernel**. The **Fejér kernel** [31] is the *average* of Dirichlet kernels. It is known to be *always positive* (no negative sidelobes) and better for convergence, but it has a *broader* main lobe, effectively smoothing the signal.

This reveals a sophisticated, **multi-resolution scan** strategy.[32] The researcher plans to first use the *smooth, broad, and stable* Fejér kernel for a *coarse, low-resolution* scan to find large "regions of interest" while avoiding false positives from sidelobes. Then, they will use the

sharp, precise Dirichlet kernel for a *fine, high-resolution* "snipe" *only inside* those promising regions. This is a standard and robust technique for efficiently searching a large signal space.

5. Critical Prognosis: Scalability and Computational Complexity

The 127-bit success is a "toy problem." The algorithm's viability depends entirely on its scaling against cryptographically relevant L-bit numbers (e.g., $L=2048$). The researcher's own "scaling hurdles" provide the necessary data to determine the computational complexity class.

5.1 The Three Stated Hurdles (and their True Meaning)

- Hurdle 1: "Peak Tightening"
  This is the researcher's observation that as N grows, the resonance peaks in the (k,m) space become narrower.34 This is a direct, practical consequence of the Nyquist-Shannon sampling theorem. As N (and thus $L=\log N$) grows, the "frequency" of the signal in the parameter space increases.35 To resolve a higher-frequency, sharper peak, the sampling density must increase.
  Given that the $\hat p(k,m)$ ansatz is an exponential function of N, the sensitivity is exponential. It is overwhelmingly likely that the peak width (Δk) scales inversely with N or a sub-exponential of N. A realistic scaling is $\Delta k \propto N^{-c} = (e^L)^{-c} = e^{-cL}$ for some $c>0$. To guarantee hitting this peak, the number of samples in the k-sweep must be $O(1/\Delta k) = O(e^{cL})$. This "peak tightening" hurdle is the exponential barrier that defines the algorithm's complexity.
- Hurdle 2: "Sensitivity Explosion" & "Precision Budget"
  The researcher estimates that the required precision d scales from ~200 dps (decimal places) at 127 bits to ~800 dps at 1024 bits. This implies d scales linearly with the bit-length L. This is entirely consistent with high-precision numerical methods, where distinguishing between two exponentially-close values requires precision proportional to L.37
  This has a direct impact on complexity. The computational cost of any algorithm is $TotalCost = (Num_of_Checks) \times (Cost_per_Check)$. The $Cost_per_Check$ (e.g., evaluating the $\hat p$ ansatz or a mod check) is not $O(1)$. Using fast multiplication (e.g., Schönhage–Strassen), the cost of a single operation at precision d is $O(d \log d \log \log d)$. Since $d \propto L$, the $Cost_per_Check$ is $O(L \log L \log \log L)$, or more simply $O(Poly(L))$.38
- Hurdle 3 (Implicit): The "Enigma of the m-Range"
  The 127-bit number was factored with $m \in [-180, +180]$. This range is tiny and, crucially, constant. This cannot hold for a general factorization algorithm.

From the ansatz derivation in Section 2.2, we found $m \approx (-k/\pi) \cdot \ln(p/q)$. For an RSA-style number (as the 127-bit example likely was), the factors are balanced: $p \approx q$. In this case, $\ln(p/q) \approx 0$, so m is small. The range $[-180, +180]$ worked for this specific case.

However, a general algorithm must factor unbalanced primes. For an L-bit N, the worst case is $p \approx 2$ and $q \approx N/2$. In this scenario, $\ln(p/q) \approx \ln(4/N) \approx -\ln N = -L \cdot \ln(2)$.

Therefore, to guarantee finding the factors of any L-bit semiprime, the m-range must scale linearly with L. The $Size_of_m_Range = O(L)$.

5.2 Likely Computational Complexity Class: EXP

We can now assemble the final complexity calculation for the WSGR "Wide-Scan" method:
$TotalCost = (Size_of_k_Range) \times (Size_of_m_Range) \times (Cost_per_Check)$

- **$Size_of_k_Range$:** $O(e^{cL})$ (Exponential, due to "peak tightening").
- **$Size_of_m_Range$:** $O(L)$ (Polynomial, due to the m-range enigma).
- **$Cost_per_Check$:** $O(Poly(L))$ (Polynomial, due to the "precision budget").

The final complexity is $TotalCost = O(e^{cL}) \times O(L) \times O(Poly(L)) = O(e^{c'L})$.
This is the definition of **Exponential Time (EXP)**. It is computationally *worse* than the **Sub-Exponential (SUBEXP)** complexity of the General Number Field Sieve.[4]

5.3 Assessment of the "Scaling Lab" Plan

The researcher's "scaling lab" plan (two-stage kernels, multi-resolution scans, consensus checks) is an attempt to *mitigate* this exponential scaling.

- **Multi-Resolution Scans:** This is a *heuristic* technique, standard in fields like robotics for "scan matching".[32] The (Fejér) scan prunes the search space for the fine (Dirichlet) scan. This is a practical optimization, but it is *not guaranteed*. It can fail if the true peak is aliased or "smoothed away" at the coarse-resolution level. It is a gamble to *mitigate* $O(e^{cL})$, not a solution that *removes* it.
- **Consensus Checks:** This term, borrowed from distributed computing [42] and AI evaluation [44], implies *parallelization* and *validation*. The researcher likely plans to run *multiple* scans in parallel (e.g., with different QMC samplers) and use a "consensus" vote to validate a candidate before spending the *extremely* expensive high-precision mod check. This is robust engineering, but it does not change the underlying $O(e^{cL})$ complexity.

The "scaling lab" plan is intelligent and employs standard engineering heuristics. However, these are *patches* applied to a *fundamentally exponential* problem. They cannot change the algorithm's complexity class.

6. Conclusion: A Novel Framework with Severe Practical Limitations

The "Wide-Scan Geometric Resonance" method is a testament to the power of *analogy* in algorithmic development. The researcher has constructed a complete factorization pipeline from a *physical metaphor* [3], a *postulated ansatz* (Section 2.2), and a *standard DSP/QMC toolkit*.[22] The 127-bit success is a significant *proof-of-concept* for this specific analogy.
The method's novel contributions are:

1. **A Novel Framework:** The core idea of transmuting factorization into a 2D DSP peak-finding problem is, to our knowledge, unique.
2. **A Bespoke *Ansatz*:** The equation $\hat p(k,m)=\exp!((\ln N-2\pi m/k)/2)$ is the "secret sauce." Its empirical success on a 127-bit number is the most significant finding and warrants further *mathematical* investigation to determine *why* and *when* this *ansatz* holds.

Despite this originality, the method is fatally undermined by its scaling properties, which are **Exponential (EXP)**.

- The primary antagonist is **"peak tightening"**, which forces an exponential increase ($O(e^{cL})$) in the sampling density of the k-dimension.
- The secondary antagonist is the **m-range**, which *must* scale polynomially ($O(L)$) to ensure generality, invalidating the hope of a constant-time search in that dimension.
- The tertiary cost is the **"precision budget"**, which makes *every* check in this massive space computationally expensive ($O(Poly(L))$).[38]

The final verdict is that the "Wide-Scan Geometric Resonance" method is **not a cryptographic threat**. It is a complex and highly original *numerical curiosity* that is ultimately defeated by the same exponential wall that has protected classical cryptography for decades. Its scaling is *worse* than the existing classical state-of-the-art (GNFS) [4], and it is not in the same universe as the polynomial-time quantum threat (Shor's Algorithm).[5]

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